

Reelmind Project — Handoff Summary

Context for continuing this project in a new chat.

What the project is

A website called **Reelmind**: the user uploads text, a PDF, or pastes a concept. The site reads/analyzes the content, breaks it into a storyboard of scenes, generates an animated video (Canvas-based visuals matched to each scene's content), narrates each scene in a chosen voice using real text-to-speech audio, and lets the user download the finished video with audio baked in. Voice options were originally meant to include male/female/boy/girl/robot; this was replaced with a real list of ElevenLabs voices plus a genuine robotic audio effect (ring modulation).

Current architecture

The project is a single self-contained HTML/JS file (**explainer-studio.html**) plus a **Cloudflare Worker** that acts as a secure backend proxy. The Worker is necessary because browsers cannot call ElevenLabs' or Anthropic's APIs directly (CORS restrictions), and because API keys must not live in client-side JavaScript.

Key names / identifiers to reuse

- Cloudflare Worker name: **plain-heart-8599**
- Worker URL: **https://plain-heart-8599.sumanshetty1722.workers.dev**
- Worker code file: **reelmind-proxy-worker.js** (has /voices, /tts/<voiceld>, /analyze, /debug routes)
- App file: **explainer-studio.html** (downloaded to C:/Users/Suman/Downloads/)
- ElevenLabs account: sumanshetty1722@gmail.com, workspace has auto-disable-if-leaked security policy

What each Worker route does

- **GET /voices** — proxies to ElevenLabs, returns available voices, using secret ELEVENLABS_API_KEY.
- **POST /tts/<voiceld>** — proxies to ElevenLabs text-to-speech, returns real audio (mp3) for one scene's narration.
- **POST /analyze** — takes the uploaded document text and asks an AI model to return a JSON storyboard (title + scenes, each with narration, visual type, and labels). Originally used Anthropic's API (needed a paid ANTHROPIC_API_KEY) — this was switched to **Cloudflare Workers AI** instead, which is free (10,000 neurons/day, no card needed). This uses a Workers AI *binding* (not a secret) named **AI**, calling model **@cf/meta/llama-3.3-70b-instruct-fp8-fast** (the original model, llama-3.1-8b-instruct, was deprecated and had to be swapped out).

- **GET /debug** — a temporary diagnostic route added during troubleshooting; shows the length/prefix/suffix of stored secrets without exposing them fully. Should be removed once everything is confirmed working.

How the audio/video pipeline works (client-side, in explainer-studio.html)

- 1. User uploads a PDF/text or pastes text.
- 2. App calls the Worker's /analyze route to get a scene-by-scene storyboard as JSON (title, narration per scene, a visual type like flow/cycle/compare/stat/concept/timeline, and short labels).
- 3. Storyboard is rendered as a filmstrip of scene cards, each with a small live Canvas preview drawn from the scene's visual type and labels (not templated per topic — genuinely driven by the AI's output).
- 4. In the Studio step, the user connects the Worker, sees a real list of ElevenLabs voices, and picks one (optionally toggling a ring-modulated 'robotic' effect).
- 5. On Generate Video: the app fetches real narration audio (via Worker's /tts route) for every scene ahead of time, decodes each into a Web Audio AudioBuffer, then plays them back in real time through a MediaStreamAudioDestinationNode combined with the Canvas's captureStream() video track, recorded together via MediaRecorder. This guarantees audio is present in the final video — no browser permission dialogs, no risk of a silent export (an earlier tab-audio-capture approach was scrapped specifically because it was unreliable).
- 6. Finished video (webm, audio included) is offered as a download.

Major problems solved during setup (long troubleshooting arc)

- ElevenLabs API cannot be called directly from a browser (CORS) — fixed by routing through the Cloudflare Worker.
- Anthropic's API likewise cannot be called directly from a downloaded/standalone HTML file outside Claude.ai's sandbox — same Worker-proxy fix, via a dedicated /analyze route.
- API keys were accidentally pasted into this chat multiple times (both ElevenLabs and, briefly, an early Anthropic path) — flagged each time and keys were rotated. IMPORTANT: never paste real API keys into chat. Set secrets only via the Cloudflare Worker's Settings, or better, via the Wrangler CLI.
- Cloudflare's dashboard UI for editing Worker secrets was silently failing to save (a known Cloudflare dashboard bug) — worked around by installing Node.js and using the Wrangler CLI instead: ``npx wrangler login``, then ``npx wrangler versions secret put <NAME> --name plain-heart-8599``, then ``npx wrangler versions deploy --name plain-heart-8599`` (select the newest version, 100% traffic, skip message). This is the reliable way to set/update secrets on this Worker going forward.

- The user did not want to pay for the Anthropic API — the whole /analyze pipeline was switched over to the free Cloudflare Workers AI binding instead, removing the need for ANTHROPIC_API_KEY entirely.
- The free Workers AI model originally chosen (@cf/meta/llama-3.1-8b-instruct) turned out to be deprecated (error 5028) — swapped to @cf/meta/llama-3.3-70b-instruct-fp8-fast.

Current status / what to check first in the new chat

- ELEVENLABS_API_KEY secret: set successfully via Wrangler CLI, confirmed working (/voices returns real voice data).
- Workers AI binding named AI: added via Cloudflare dashboard (Settings/Bindings tab), confirmed connected.
- Worker code: was just updated to use the new (non-deprecated) model and to wrap /analyze in a try/catch that returns readable JSON errors instead of crashing (crashes were previously showing up misleadingly as CORS errors in the browser).
- NOT YET CONFIRMED: whether Analyze & Build Storyboard succeeds end-to-end with the new model — this was the very next test to run when this chat ended.
- STILL PENDING: one more ElevenLabs key rotation, since the key was exposed in chat a few times during troubleshooting. Once everything is confirmed working, create a fresh ElevenLabs key and set it via the Wrangler CLI method only.
- The ANTHROPIC_API_KEY secret is no longer used by the code but may still exist as a stored secret on the Worker — harmless, but could be deleted for cleanliness.
- The /debug route in the Worker should be removed once everything is verified, since it's a diagnostic-only endpoint.

What to ask the new chat to do

Share this PDF, then ask Claude to: (1) confirm the Analyze step now works with the new model, helping debug via the browser console/Network tab the same way as before if it doesn't; (2) once confirmed, do the final ElevenLabs key rotation via Wrangler CLI; (3) optionally clean up the unused ANTHROPIC_API_KEY secret and the temporary /debug route; (4) continue polishing the app itself (e.g. richer scene animations, better mobile layout, etc.) if desired.